

Git Tags – Complete Guide (Final Corrected)

1. What is a Tag in Git?

A tag in Git represents a version or release of your project. It marks a specific point in history.

2. Purpose of Tags

- Track versions (v1.0, v2.0)
- Identify important commits
- Used for releases

3. Git Tag Command

`git tag`

4. Types of Tags

A. Annotated Tag

Stores metadata (author, date, message)

Command:

```
git tag -a -m "message"
```

Example:

```
git tag -a v1.0 -m "First release"
```

B. Lightweight Tag

Simple reference without metadata

Command:

```
git tag
```

Example:

```
git tag v1.0
```

5. Tag Specific Commit

```
git tag
```

6. View Tags

`git tag`

7. Push Tags to GitHub

Push single tag:

`git push origin`

Example:

`git push origin v1.0`

Push all tags:

`git push origin --tags`

Note:

- 'git push origin' alone will NOT push tags
- 'git push origin --tags' pushes all tags correctly

8. Key Difference

Annotated → full details

Lightweight → simple label

9. Summary

Use annotated tags for releases and lightweight tags for quick marking.